UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION (SEMESTER I) – 2019/2020
CS101 - INTRODUCTION TO COMPUTER SCIENCE (3 CREDITS)

Answer **ALL** Questions Time Allowed: **TWO Hours**

Registration No: _ _ _ _ _ _ _ _ _ _ _ _ _ _

Instructions to Candidates

1) Check the number of pages on the question paper before answering.
2) Answer all questions in the space provided.
3) Write your registration number clearly in all indicated pages.
4) Marks distribution for each question is follows. You may manage your time accordingly.

Q1	Q2	Q3	Q4	Q5
30	15	15	25	15

Question - 01

1.1 What is a non-positional number system?

[02 Marks]

1.2. Give two examples each for positional and non-positional number systems.

	Positional Number Systems	Non-positional Number Systems
i		
ii		

[04 Marks]

1.3 *RGB* (Red, Green, Blue) is a colour scheme used for mixing color in computer displays. The *RGB* colours are usually given in hexa-decimal form. Each component (R, G and B) has two hexa-decimal characters (example: *34A1DE*, Red: 34, Green: *A1* and Blue: *DE*).

Consider the given RGB colour is *63CCA2* and fill the following table.

Calculations:

[*Continue 1.3*]

	Decimal Representation	Binary Representation
Red		
Green		
Blue		

[06 Marks]

1.4 The following structure is about single precision IEEE 754 floating point number representation.

[A] bit/s		
B	C	D
[E] bit/s	[F] bit/s	[G] bit/s

Fill the names / values from A to G considering the above information.

A	B	C	D	E	F	G
32	sign	exponent	mantissa	1	8	23

[07 Marks]

1.5 Using single precision IEEE 754 format, convert the decimal number -6.625 and show the answer in Binary representation. You should clearly show all calculations.

Calculations:

$6 \rightarrow 110$

0.625×2 1.250 — 1

0.250×2 0.5 0

0.5×2 1.0 1

101

110.101×10^2

1.10101

$127 + 2 = 129$

$2 \,|\, 129$
$2 \,|\, 64 - 1$
$2 \,|\, 32 - 0$
$2 \,|\, 16 - 0$
$2 \,|\, 8 - 0$
$2 \,|\, 4 - 0$
$2 \,|\, 2 - 0$
$1 - 0$

Binary Representation:

1	10000001	1010100000000000000000 0000

[11 Marks]

Question - 02

2.1 What is meant by arithmetic overflow? [02 Marks]

2.2 Illustrate arithmetic overflow using a suitable example. [03 Marks]

2.3 State two main reasons to use 2's complements instead of sign-magnitude in high-level programming languages?

i) has a single representation for zero

ii) +, − work the same for both + & − numbers.

[04 Marks]

2.4 Critically evaluate the following three snippets (code blocks) in terms of CPU usage while executing the instructions.

a) $2 + 2$ b) 2×2 c) 2.0×2.0

[06 Marks]

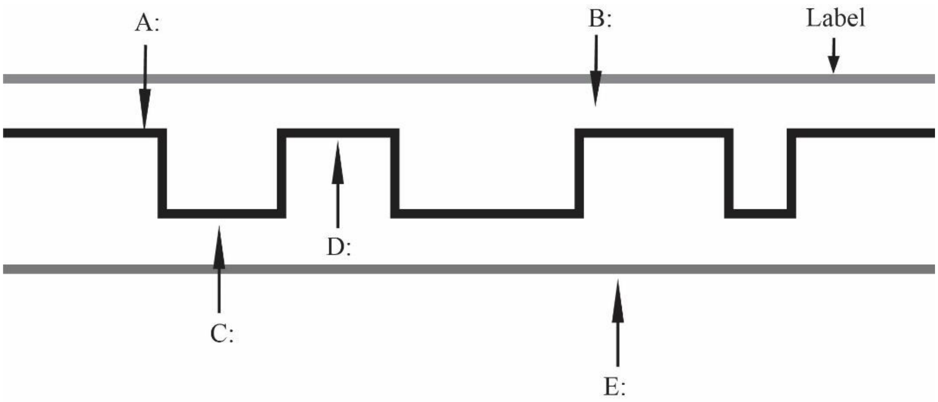
Question - 03

3.1 Fill the following table comparing magnetic disk drives and solid-state drives.

	Property	Magnetic Disk Drives	Solid-State Drives
i	Operating Speed		
ii	Durability		
iii	Power Consumption		
iv	Price		

[04 Marks]

3.2 The following structure shows a cross section of an optical disk. Label the components from A to E using Acrylic, Aluminum, Land, Pit and Polycarbonate plastic.



[05 Marks]

3.3 Explain how "Blu-ray" discs hold more data than DVDs.

[02 Marks]

3.4 What does NVMe stand for:

[01 Marks]

3.5 The following statement is about SRAM and DRAM. Fill the blanks using the options given within parenthesis. Circle the correct answer.

"*SRAM* is an …… (off-chip / on-chip) memory whose access time is …… (small / large) while *DRAM* is an …… (off-chip / off-chip) memory which has a …… (small / large) access time. Therefore, *SRAM* is …… (slower / faster) than *DRAM*." [03 Marks]

Registration No: _ _ _ _ _ _ _ _ _ _ _ _ _ _

Question - 04

4.1 What is a proposition?

[02 Marks]

4.2 Write two non-proposition examples.

i)

ii)

[02 Marks]

4.3 Is the following 4-variable (A,B,C, D) Karnaugh map valid? Justify your answer.

AB

CD	**10**	**00**	**01**	**11**
10				
00				
01				
11				

Validity: Yes / No (circle your answer)

Justify your answer:

[04 Marks]

4.4 State three main reasons to simplify Boolean expressions.

i)

ii)

iii)

[03 Marks]

4.5 Simplify the following 3-variable Boolean function using a Karnaugh map.

$$f(X,Y,Z) = \bar{X}YZ + \bar{X}Y\bar{Z} + XZ$$

[02 Marks]

4.6 Prove the following equality using standard Boolean algebra theorems. You must clearly state the theorem that you used at each step.

$$XY + \bar{X}Z + YZ = XY + \bar{X}Z$$

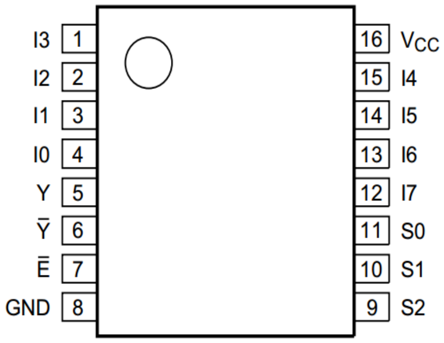
[04 Marks]

4.7 Implement the following Boolean function with an 8×1 multiplexer and minimum external gates.

$$F(A, B, C, D) = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$$

[06 Marks]

4.8 Illustrate your answer for 4.7) using the 8×1 Mux diagram below. You must clearly show inputs, outputs, external gates and power source connections.



[02 Marks]

Question - 05

5.1 State three main characteristics of an algorithm.

i :

ii :

iii :

[03 Marks]

5.2 State two main advantages of pseudocodes over flowcharts.

i :

ii :

[04 Marks]

5.3 Develop a flow chart to display all the numbers having '9' digit for given boundaries (lower and upper limits, both inclusive). The lower and upper limits should be given as keyboard inputs.

Example:

Input: lower = 1, upper = 100

Output: 9, 19, 29, 39, 49, 59, 69, 79, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99

[08 Marks]